

Chap 4: Intellectual Property

- o Principles, Laws, and Cases
- o What Is Intellectual Property?
- o Challenges of New Technologies
- o A Bit of History
- o Ethical Arguments About Copying
- o Significant Legal Cases
- o Search Engines and Online Libraries
- o Free Software
- o What Is Free Software?
- o Should All Software Be Free?
- o Search Engines and Online Libraries
- o Free Software
- o What Is Free Software?
- o Should All Software Be Free?

What is **Intellectual Property**?



Intellectual property refers to the legal rights protecting creative works and inventions, including literary works, music, designs, trademarks, and trade names.

These rights grant the creator or owner exclusive control over the use and distribution of their work, preventing others from copying or using it without permission.

Intellectual property is protected through copyright laws, patents, trademarks, and other legal mechanisms.

This fosters innovation, creativity, and recognition of creators' rights.

Core Principles of Intellectual Property (IP)

Core Principles of Intellectual Property (IP)

Intellectual Property laws are designed to:

- o **Encourage innovation and creativity**
- o **Reward creators financially**
- o **Protect ownership rights**
- o **Balance public access and private rights**

IP mainly includes:

- o **Copyright** – protects books, software, music, videos
- o **Patents** – protects inventions and processes
- o **Trademarks** – protects brand names and logos
- o **Trade Secrets** – protects confidential business information

What Is Intellectual Property?

Intellectual Property (IP) refers to creations of the mind such as:

- o Software code
- o Research papers
- o Digital art
- o Music
- o Designs
- o Algorithms

In Computer Science, IP covers:

- o Source code , Apps, Database, Websites, AI models

Example:

- o The Windows operating system is protected by copyright owned by Microsoft.
- o The iPhone design and technology are protected by patents owned by Apple Inc..

Challenges of New Technologies

Challenges of New Technologies

- o Technology creates new IP problems:

Digital Copying

- o Easy to copy software, movies, books.
- o One click = unlimited duplication.

Internet Distribution

- o Peer-to-peer sharing (e.g., torrenting).
- o Cloud storage sharing.

AI & Content Creation

- o Who owns AI-generated content?
- o Is training AI on copyrighted data legal?

3D Printing

- o Physical products can be copied digitally.

Example:

Music-sharing platforms like Napster created major copyright conflicts.

A Bit of History

Early Copyright Law

- o First copyright law: **Statute of Anne (1710)** in England.
- o US Copyright Act: 1790.

Software Protection

- o In the 1970s, software was not clearly protected.
- o Later recognized as copyright-protected.

Digital Millennium Copyright Act (DMCA) – 1998

- o Protects digital content.
- o Criminalizes bypassing DRM (Digital Rights Management)

Ethical Arguments About Copying

Arguments Against Copying (Piracy is Wrong)

- o Creators deserve payment.
- o Copying harms innovation.
- o It is theft of intellectual labor.

Arguments Supporting Copying

- o Information should be free.
- o High prices limit education.
- o Sharing promotes knowledge.

Ethical debate:

Is downloading a movie illegally the same as stealing a physical object?

Significant Legal Cases

A&M Records v. Napster (2001)

- Napster was sued for enabling music piracy. Court ruled Napster responsible for copyright infringement.

MGM Studios v. Grokster (2005)

- MGM Studios sued Grokster. Supreme Court ruled companies encouraging piracy can be liable.

Authors Guild v. Google (2015)

- Authors Guild sued Google over Google Books scanning. Court ruled Google's scanning was **fair use**.

Search Engines and Online Libraries

Search Engines

Search engines like Google:

- o Copy web pages temporarily.
- o Store cached versions.
- o Display snippets.

Legal issue:

Is copying web pages for indexing copyright violation? Courts generally allow it under **fair use**.

Online Libraries

Examples:

- o Project Gutenberg (public domain books)
- o Internet Archive

Issues:

- o Digitizing copyrighted books
- o Controlled digital lending
- o Publisher lawsuits

Free Software

What Is Free Software?

- o Free software means:
- o Freedom to use
- o Freedom to modify
- o Freedom to share
- o Not necessarily "zero price".

Founded by: Richard Stallman

Organization: Free Software Foundation

Example:

- o Linux
- o GNU Compiler Collection

Key Freedoms (Four Freedoms)

- o Run the program
- o Study the code
- o Modify it
- o Share copies

Should All Software Be Free?

Arguments FOR Free Software

- Promotes innovation
- Reduces monopoly
- Improves security (code is open)
- Encourages collaboration

Example:

Linux powers:

- Servers
- Android phones
- Supercomputers

Arguments AGAINST All Software Being Free

- Developers need financial incentive.
- Companies invest billions in R&D.
- Some projects require strong commercial protection.

Example:

Microsoft sells Windows as proprietary software for revenue.

□ Comparison: Free vs Proprietary Software

Feature	Free Software	Proprietary Software
Source Code	Open	Closed
Modification	Allowed	Restricted
Cost	Often free	Paid
Control	User	Company